Per-unit and synchronous reactance

Practice questions · Day 13 · Thu 5 Nov · pack M10

5. Practice

From your course files

PQ 8.1 (practice questions). For a single-phase system calculate the following assuming a base voltage of 240V and base power of 2000VA.

(a) base current
(b) base impedance
(c) base resistance
(d) base reactance

PQ 8.2 (practice questions). Using the results from the previous question, solve Question 2.17 using the per-unit method, for parts (a) -(c) and (e) give the answers in pu, for parts (f)-(h) convert the results back to real units

Question 2.17 (for reference): A single-phase load consists of a parallel-connected 25 ohm resistor and an inductor with a 125 ohm reactance is connected to the 240 V mains, calculate
(a) the load impedance Z in polar form
(b) write the supply voltage $\mathbf{V}_1$ in phasor form (assume it has a phase angle of zero)
(c) the resultant phase current phasor $\mathbf{I}_1$
(d) the input power factor (remember to show lead or lag)
(e) the complex power $\mathbf{S}_1$ in both polar and Cartesian form
(f) the input real power P using
  i) the real part of $\mathbf{S}_1$
  ii) the input power factor and V_1 and I_1
  iii) the fact that all the power losses occur in the resistor and hence the losses are V^2/R (why not I^2R?)
(g) the input reactive power Q
(h) the input VA (that is, the apparent power S)
(i) sketch a power triangle and verify that $P^2 + Q^2 = S^2$
(j) sketch a phasor diagram showing the relationship between the voltage and the current

PQ 8.3 (practice questions). Repeat the calculations in Question 2.20 Part (a) only to find the magnitude of the two currents using the per-unit system. Use a base VA of 1000VA and base voltage of 200V.

Question 2.20 (for reference): Two single-phase loads $Z_1 = 100 + j0\ \Omega$ and $Z_2 = 10 + j20\ \Omega$ are connected in parallel across a 200 V, 60 Hz source.
(a) Find the magnitude of the two load currents, the total real and reactive power supplied by the source, and the power-factor at the source.

PQ 8.4 (practice questions). A calculation is to be carried out using the per-unit system to a base of 10 MVA. It relates to a section of a (three-phase) power supply system whose nominal line voltage is 11 kV. Calculate the following quantities:

(a) The base line voltage
(b) The base phase voltage
(c) The base phase current
(d) The base phase impedance
(e) The base phase reactance
(f) The base phase resistance
(g) The base real power
(h) The base reactive power
(i) The base VA

PQ 8.5 (practice questions). Express the following as impedances in a three-phase system in per unit form to a base of 750 kVA :

(a) 5 ohm inductive reactance on a 3ph 3300 volt distribution line.
(b) 0.05 ohm resistance on a 3ph 415V supply

PQ 8.6 (practice questions). A three-phase synchronous generator has a rated output voltage of 415 V. Under a particular operating condition it can be represented by an internal voltage source of (line) voltage 440 V and series phase reactance of 7.5 ohms. Carry out the following calculations using a base of 10 kVA.

(a) Calculate the base current and base impedance.
(b) Draw the equivalent circuit for the machine using per-unit values for the internal voltage and impedance.
(c) Calculate the magnitude of the short-circuit output current in per-unit. Then convert this value to amps.
(d) Check this answer by performing the calculations using conventional units.

PQ 8.7 (practice questions). Calculate the magnitude and phase of the per-unit stator voltage $\mathbf{V}$ and current $\mathbf{I}$ for the following situations. You can assume the phase of the voltage is taken as the reference. Use the generator rating as the power base. [Hint : use the relations $I = S/V$ pu and the power triangle relating S, P and Q, and also that the angle of $\mathbf{I}$ equals the power factor angle $= \cos^{-1}(\mathrm{pf})$. Watch the sign of the current angle depending on whether the power-factor is leading or lagging.]

(a) A synchronous generator is operating at full load (rated VA, i.e. S=1pu) at a power-factor of 0.9 lagging. The terminal voltage is at its rated value.
(b) A synchronous generator is operating at half load (that is, half rated output VA, S=0.5pu) at a power-factor of 0.9 leading. The terminal voltage is 5% above its rated value.
(c) A 150 MVA generator is producing 100 MW at a power-factor of 0.85 lagging. The terminal voltage is at its rated value.
(d) A 200 MVA generator is producing 150 MW and 50 MVAr lagging. The terminal voltage is 5% below its rated value

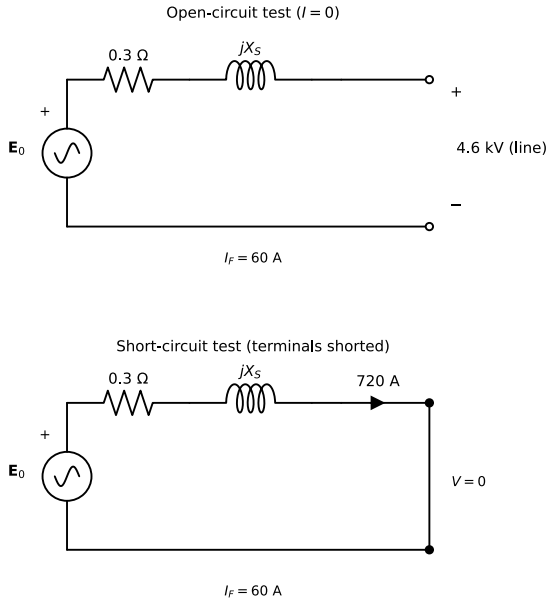
Jun23 Q4a (past exam, 5 marks). *4a) Synchronous Machines [5 marks]* A 150 MVA, 50Hz, 3,000rpm, 14 kV power station generator gives the following test results:

Excitation (amps)	Open-circuit line voltage (kV)	Short-circuit current (A)
60	4.6	720

The stator resistance is measured at 0.3 ohm.

i) Find the value of synchronous reactance in ohms (not per-unit). [2 marks]

ii) The generator is delivering a real power of 100MW and reactive power of 50 MVAr at rated voltage. Calculate the magnitude and phase of the stator voltage and current in per-unit. [3 marks]

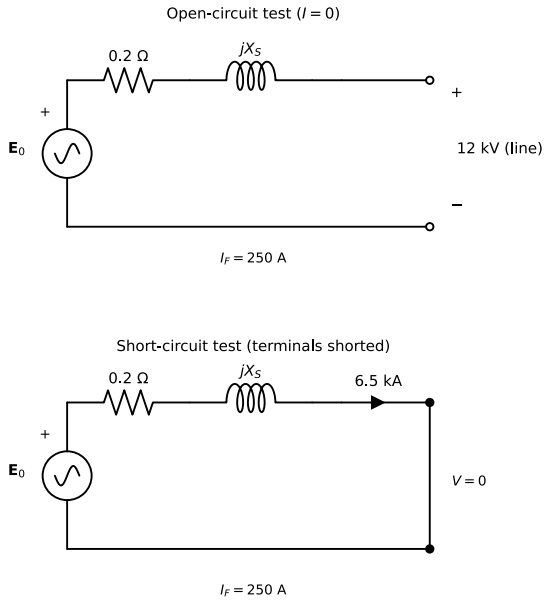


Jun23 Q4a test circuits: OC 4.6 kV line and SC 720 A, both at a field current of 60 A, R = 0.3 ohm

Jun24 Q4a (past exam, 5 marks). *4a) Synchronous Machines [5 marks]* A 14 kV, 400 MVA synchronous generator is tested at a field current of 250 A. The open-circuit voltage was found to be 12 kV and the short-circuit current was 6.5 kA. The stator resistance was measured at 0.2 ohms per phase.

i) Find the value of the synchronous reactance in per-unit. [3 marks]

ii) The synchronous generator is delivering 200MW at a terminal voltage of 3% below rated voltage and a power-factor of 0.8 lagging. Find generator output voltage and current phasors in per-unit. [2 marks]



Jun24 Q4a test circuits: OC 12 kV line and SC 6.5 kA, both at a field current of 250 A, R = 0.2 ohm

PQ 8.11 (practice questions; parts (a)–(b) here). A power station generator is rated at 250 MVA, 15 kV and has a star-connected stator winding. After manufacture, an open-circuit test is performed and a line voltage of 5.75 kV is measured with an excitation of 50 A. Under the same value of excitation, a current of 2050 A is measured with the short-circuit test. The resistance of each stator phase is 0.15 ohm. The generator is supplying its full-load rating at nominal voltage and a power factor of 0.85 lag. Ignoring the effects of magnetic saturation, calculate :

(a) The unsaturated synchronous reactance in per-unit
(b) The per-unit stator current magnitude and phase relative to the terminal voltage
(c) The induced voltage E_0 in per-unit showing both its magnitude and phase angle [Hint : draw the synchronous machine equivalent circuit and hence see that $E_0 = V + I(R + jX_S)$]
(d) The corresponding value of E_0 as a line voltage. [Hint : multiply the magnitude of the per-unit value by the base line voltage]
(e) The excitation required in amps.
(f) The rotor angle?

PQ 8.12 (practice questions; parts (a)–(b) here). A 120 MVA, 50Hz, 3,000rpm, 16 kV power station generator gives the following test results :

Excitation (amps)	Open-circuit line voltage (kV)	Short-circuit current (A)
50	4.4	600

The stator resistance is measured as 0.1 ohm. Magnetic saturation can be ignored.

(a) From the test results, calculate the values of the unsaturated synchronous reactance in per unit
(b) Calculate the stator current magnitude and phase in per-unit when the generator is supplying a load of 102 MW at a power factor 0.95 lag when the terminal voltage is at its rated value. Use the terminal voltage as the reference phasor.
(c) Calculate the required excitation (in amps) for the situation in part (b). What is the ratio of this value of excitation current compared to the excitation current required for synchronization?

New questions

New Question 1 (easy) [4 marks]. A 3ph system has a base of 50 MVA and 11 kV. (a) Find I_B and Z_B. [2] (b) Express a 0.8 Ω per-phase reactance in pu. [1] (c) A bus voltage is 0.95 pu. Give it as a line voltage and as a phase voltage in kV. [1]

New Question 2 (easy–medium) [5 marks]. A 300 MVA, 18 kV, star-connected generator is tested. The OC slope on the linear part is 400 V (line) per field amp, and the SC slope is 80 A per field amp. $R = 0.05\ \Omega$. (a) Find X_S in ohms and in pu. [3] (b) What field current gives rated voltage on open circuit? [1] (c) What field current gives E_0 = 1.8 pu, assuming no saturation? [1]

New Question 3 (medium) [4 marks]. A 250 MVA, 15 kV generator delivers 180 MW and **absorbs** 60 MVAr from the grid, with the terminal voltage 3% above rated. Find $\mathbf{V}$ and $\mathbf{I}$ in pu, and state the pf.

New Question 4 (exam-style, chained) [9 marks]. A 180 MVA, 15 kV, star-connected generator gives these test results at I_F = 80 A: OC line voltage 6.0 kV, SC current 2400 A. $R = 0.12\ \Omega$. Ignore saturation.

a. Find X_S and R in pu. [3]
b. The generator delivers 135 MW at 0.9 pf lagging with the terminal voltage 2% above rated. Find $\mathbf{V}$ and $\mathbf{I}$ in pu, and I in amps. [3]
c. Find $\mathbf{E}_0$ in pu, E_0 as a line voltage, and the field current required. [3]

New Question 5 (motor convention) [5 marks]. A 500 kVA, 415 V synchronous motor has $X_S = 0.9$ pu, with R neglected. It draws 400 kW at 0.8 pf **lagging** at rated voltage. (a) Find the input current in pu and in amps. [2] (b) Using the motor convention (current into the machine), find $\mathbf{E}_0$ in pu and as a line voltage. [2] (c) Is the rotor angle positive or negative, and what does that tell you? [1]

6. Quick recall check

1. A 3ph machine is rated 120 MVA, 16 kV. What is Z_B?

2. The OC test gives 4.4 kV (line) at I_F = 50 A, and the SC test gives 600 A at the same I_F. What is $|Z_S|$ in ohms?
3. A generator supplies 0.6 pu real power at 0.8 pf leading, at 1.0 pu voltage. What is $\mathbf{I}$?
4. Write the course's equivalent-circuit equation for a generator, and the motor version using the input current $\mathbf{I}_m$.
5. Why does the short-circuit characteristic stay linear when the open-circuit characteristic saturates?
